

AAIS Commentary Codex

Tri-Model Analysis & Architectural Insights

Version	1.0
Date	April 2026
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Contributing Models	Copilot (Microsoft) · Grok (xAI) · Claude (Anthropic)
Reference Specification	AAIS_MASTER_SPEC v1.2
Classification	Internal — Architectural Commentary & Analysis

Purpose

This codex captures the architectural commentary, formalization insights, logic-pressure corrections, and design observations produced during tri-model review of the AAIS specification. It is not a consensus document — it preserves each model's analytical perspective while surfacing convergences and productive tensions.

1. How This Document Works

Each major AAIS architectural domain is examined through three analytical lenses. Commentary is attributed by model. Where all three models converge on a conclusion, a **Convergence Note** captures the shared assessment. Where they

diverge — or where a model identifies an unresolved design question the others did not raise — a **Tension Note** surfaces the disagreement for operator review.

This is a living document. As the AAIS architecture evolves, new commentary layers can be appended without disrupting prior analysis. Each section is self-contained: readers may enter at any architectural domain without losing context.

1.1 Commentary Format

Each section follows a consistent pattern:

COPILOT Perspective — Practical engineering analysis, integration considerations, implementation feasibility

GROK Perspective — Formalization and structural rigor, mathematical/logical grounding, edge-case pressure

CLAUDE Perspective — Logic-pressure testing, philosophical coherence, failure-mode analysis, governance completeness

Convergence Notes — Where all three models agree

Tension Notes — Where models disagree or identify unresolved design questions

2. Commentary on System Philosophy

2.1 "Governed Cognition Over Autonomous Improvisation"

COPILOT

This principle is architecturally sound but requires careful implementation to avoid over-constraining adaptive subsystems. The key tension is between governance overhead and cognitive throughput. Recommendation: governance should be **lightweight at the packet level** and **heavy at the boundary level**. Per-packet governance checks should be fast lookups against pre-compiled rule tables. Boundary governance — cross-module transitions, doctrine activation, identity assertions — can afford deeper evaluation because it occurs less frequently.

GROK

Formalizing this principle requires a clear delineation between "governed" (deterministic rule application) and "cognition" (probabilistic inference). The spec should define the formal boundary where governance yields to inference and vice versa. Without this, "governed cognition" risks becoming a slogan rather than an enforceable property. Specifically, the spec needs a decision function $D(x)$ that, for any system state x , returns whether the current operation is in governed mode or cognitive mode — and the transition conditions between them must be well-defined.

CLAUDE

The deeper question is whether governance and cognition are truly separable at runtime. In practice, the act of governing introduces its own cognitive overhead and decision-making. The spec should acknowledge that the governance layer itself requires governance — a **meta-governance problem**. The Cognitive Bridge Module partially addresses this but doesn't fully close the recursion. If the governance layer makes a probabilistic judgment about whether a subsystem has drifted, that judgment is itself a cognitive act — subject to error, bias, and drift. The architecture needs a termination condition for this recursion.

✓ **Convergence Note**

All three models agree this is the correct foundational principle for a system of this class. No model suggests autonomy-first or ungoverned cognition as a viable alternative for AAIS.

⚠ **Tension Note**

Disagreement on whether the governance-cognition boundary can be made **formally clean** (Grok's position) or whether it will always require **operator judgment at the margin** (Claude's position). Copilot takes a pragmatic middle ground: make it clean where possible, accept judgment where necessary, and instrument both.

2.2 Operator Sovereignty

COPILOT

Operator sovereignty is well-defined at the Jarvis level but the spec should clarify what happens when operator intent conflicts with system law. Does the operator override law, or does law constrain the operator? This is not an edge case — it will arise routinely during development and testing. The spec needs an explicit precedence chain: *Law > Operator > Doctrine > Subsystem Autonomy*, or whatever the intended ordering is.

GROK

This is a classic authority paradox. The spec implicitly treats the operator as sovereign over Jarvis, and Jarvis as sovereign over subsystems. But law is described as immutable. If the operator issues a command that violates law, the

system must choose — and the spec should explicitly state the resolution order. Formally: let O = operator command, L = system law. If $O \cap \neg L \neq \emptyset$, what is the resolution? Reject O ? Constrain O to $O \cap L$? Log the conflict and defer?

CLAUDE

The Identity Invocation Lane is a good example of operator sovereignty working correctly — direct address preempts all other lanes. But this model breaks down under adversarial or confused operator inputs. What prevents an operator from inadvertently triggering IIL during a legitimate debug session? The negative filter (operator-task keywords) is a pragmatic fix but not a principled one. A principled solution would define IIL triggering as a two-stage confirmation: intent detection + operator confirmation before lane preemption occurs.

3. Commentary on Architecture

3.1 Direct Internal Pipeline

COPILLOT

The fast-lane/slow-lane separation is architecturally clean. The packet model is well-specified. Implementation concern: the packet schema needs **versioning from day one**. As the system evolves, packet formats will change, and backward compatibility must be governed. Each packet should carry a schema version field, and the pipeline should include version-negotiation logic at module boundaries. Without this, schema evolution will become a source of silent failures.

GROK

The pipeline formalization is strong but incomplete. The spec defines packet fields but doesn't define **packet ordering guarantees, delivery semantics** (at-most-once, at-least-once, exactly-once), or what happens when a packet is dropped mid-pipeline. These are critical for a system claiming determinism. A pipeline without delivery guarantees is a pipeline without reliability proofs. The spec should specify: (1) whether the pipeline is FIFO-ordered, (2) what the retry semantics are, and (3) what constitutes a delivery acknowledgment.

CLAUDE

The "no prose transport" rule is one of the strongest architectural decisions in the spec. It prevents the most common failure mode in multi-agent systems: **context drift through natural language reinterpretation**. However, the spec should define what counts as "prose" vs. structured payload — the boundary between a compact semantic tag and a natural language description is fuzzy. A field value like *"user is asking about weather"* looks like prose but functions as a semantic tag. The spec needs a structural test for "prose-ness" that can be applied at the packet validation layer.

✓ **Convergence Note**

All three models affirm the fast-lane/slow-lane separation as architecturally correct and the packet model as the right abstraction for internal communication. The "no prose transport" rule receives universal endorsement.

⚠ **Tension Note**

Grok identifies missing delivery semantics as a **significant gap** that undermines the system's determinism claims. Claude and Copilot agree this needs attention

but differ on priority — Copilot prioritizes schema versioning; Claude prioritizes the prose/structured boundary definition.

3.2 Lane Separation & Traffic Classes

COPILOT

Traffic class separation (A, B, C) needs **monitoring infrastructure from day one**. Without observability into which traffic is flowing where, lane violations will go undetected. Recommendation: each packet should carry its traffic class as a first-class field, and the pipeline should emit class-level metrics at every routing decision point. Dashboards should show traffic distribution by class in real time.

GROK

The traffic classes should be formally defined as a **type system**, not just labels. Class A traffic should have a distinct type signature from Class C, enforced at compile time where possible and at runtime where not. This prevents misclassification at the source — a module cannot emit Class A traffic unless its output type matches the Class A schema. Type confusion between traffic classes is a category error that should be structurally impossible.

CLAUDE

The deeper risk is not lane violation but **lane starvation** — Class A traffic consuming all bandwidth and starving governance traffic (Class B). The spec should define priority inversion protections. In a system where governance depends on timely Class B delivery, allowing Class A to monopolize the pipeline is a structural failure that degrades governance precisely when governance is

most needed (high-load scenarios).

4. Commentary on Cognitive Bridge Module

4.1 Intent Resolution

COPILOT

Intent resolution is the most implementation-sensitive component. The spec defines *what* the Intent Resolver does but not *how* it determines confidence levels. A low-confidence intent resolution should trigger different behavior than a high-confidence one — and the spec should define the threshold model.

Recommended approach: a three-band model (high confidence > 0.85 , medium $0.5\text{--}0.85$, low < 0.5) with distinct routing behavior per band.

GROK

The Governed Intent Packet schema includes a "risk_class" field, which is good. But risk classification is itself a governance problem — who classifies risk, and is the classification auditable? The bridge should log not just the result but the **reasoning path** that produced the risk classification. Without audit trail on classification logic, risk_class becomes an opaque label rather than a verifiable property.

CLAUDE

The Intent Resolver is the **single most dangerous component** in the system because it sits between raw operator input and doctrine activation. If it misclassifies intent, the wrong doctrine binds, the wrong invariants activate, and the system behaves incorrectly but auditably — which is *worse* than crashing, because the error looks legitimate. A misclassified intent produces a coherent, well-governed, completely wrong response. The system will have done everything correctly except the one thing that matters most.

✓ Convergence Note

All three models identify Intent Resolution as the **highest-risk, highest-impact subsystem** in the Cognitive Bridge Module. This is the component most deserving of additional specification, testing, and operator-facing transparency.

4.2 Mythic Translation Registry

COPILOT

The registry is a strong design pattern. Implementation note: the registry should be loaded from a **configuration file**, not hardcoded, so operators can extend it without code changes. But extensions must go through a governance review process — unreviewed mythic mappings are a vector for unintended behavior. A staged pipeline (propose → review → activate) is the right model for registry changes.

GROK

The registry formalizes a mapping from a symbolic domain to an operational domain. This is essentially a controlled vocabulary with semantic bindings. The formalization should include: (1) **uniqueness constraints** — no two symbols

map to the same operation; (2) **completeness checks** — are there operational concepts with no symbolic mapping?; (3) **consistency checks** — do any mappings contradict each other? These three properties (uniqueness, completeness, consistency) are the standard requirements for any formal translation layer.

CLAUDE

The registry rule — *"No mythic symbol may affect runtime behavior unless it has an approved mapping entry"* — is the most important single sentence in the Cognitive Bridge Module specification. It creates a hard wall between narrative resonance and runtime truth. But the inverse question is equally important: **can an operational concept exist without a mythic mapping?** If yes, the system has "unmythologized" operations that may confuse operators who think in mythic terms. If no, the registry becomes a bottleneck for all new functionality. The spec must answer this question explicitly.

✓ Convergence Note

Universal agreement that the Mythic Translation Registry is architecturally necessary and well-conceived. The controlled-vocabulary pattern with governance gating is endorsed by all three models.

⚠ Tension Note

Claude raises the **inverse completeness question** — whether unmythologized operations are permitted — that neither Copilot nor Grok address. This has significant UX and scalability implications that remain unresolved.

4.3 Drift Detection

COPILLOT

Drift detection needs **quantitative metrics**, not just categorical classifications. "High severity" drift means different things in different contexts. The system needs drift scoring — a numerical measure of how far a subsystem has deviated from its identity manifest. This score should be tracked over time and visualized per-module on an operator dashboard.

GROK

The drift detection model should be formalized as a **distance function over the space of allowed behaviors**. Each subsystem has an identity manifold — a bounded region in behavior space. Drift is the distance from the current behavior point to the nearest point on the manifold boundary. This makes drift detection a geometric problem with clean formal properties: convergence guarantees, measurability, and threshold-based alerting.

CLAUDE

The hardest drift to detect is **slow drift** — gradual identity erosion that stays within tolerance on any single turn but accumulates over sessions. The spec's per-turn drift monitor won't catch this. A session-level or multi-session drift accumulator is needed. This connects to the broader question of **temporal governance** — most of the spec is turn-scoped, but real drift is trajectory-scoped. A subsystem that drifts 0.5% per turn for 200 turns has drifted 100% — but no single turn triggered an alert.

⚠️ Tension Note — Three-Way Divergence

Fundamental disagreement on the right abstraction level for drift detection.

Grok wants geometric formalization (distance functions on behavior manifolds).

Claude wants temporal accumulation (trajectory-scoped monitoring). **Copilot** wants quantitative scoring (dashboard-ready numerical metrics). All three are valid and potentially complementary — the spec should integrate all three perspectives into a layered drift detection architecture.

5. Commentary on Immune Protocol

5.1 Detection vs. Governance

COPILLOT

The separation between Lane Governor (rules) and Immune Protocol (detection + response) is clean. But the spec should define what happens when the Immune Protocol and the Lane Governor **disagree** — e.g., the Governor allows a packet but the Immune Protocol flags it as anomalous. This is not a hypothetical: any rule-based system and any pattern-based system will produce conflicting classifications on boundary cases. The spec needs a conflict resolution policy.

GROK

The Immune Protocol's classification scheme (NORMAL, ANOMALY, VIOLATION) should be formalized with clear **transition rules**. Can a NORMAL classification be downgraded to ANOMALY mid-session? Can an ANOMALY be upgraded to NORMAL if the pattern resolves? The state machine needs explicit definition. Formally: define states $S = \{N, A, V\}$, transitions $T: S \times Event \rightarrow S$, and the event

types that trigger each transition.

CLAUDE

The most powerful aspect of the Immune Protocol is that it catches "**technically legal but structurally wrong**" behavior — things that pass Governor rules but still indicate system degradation. This is analogous to how biological immune systems detect foreign patterns even when individual cells appear normal. The risk is false positives — an overly aggressive Immune Protocol will quarantine legitimate traffic. The spec should define the false positive tolerance and the recalibration procedure when false positive rates exceed threshold.

5.2 Response Proportionality

COPILLOT

The five response types (ALLOW, CLAMP, REROUTE, REJECT, QUARANTINE) need clear **escalation logic**. Under what conditions does the system escalate from CLAMP to REJECT? The spec should define escalation thresholds — including time-based escalation (CLAMP held for more than N turns triggers REROUTE) and severity-based escalation (anomaly score exceeding threshold triggers REJECT).

GROK

Response proportionality is a formal **control theory problem**. The Immune Protocol needs a feedback loop — if CLAMP doesn't resolve the anomaly within N turns, escalate to REROUTE. If REROUTE fails, REJECT. If REJECT is insufficient, QUARANTINE. This is a standard graduated response model that should be specified with concrete parameters. The parameters (thresholds, timeouts, retry

counts) should be tunable per module and per traffic class.

CLAUDE

The QUARANTINE response is the most dangerous because it can **cascade**. Quarantining a core module (e.g., Nova) affects all downstream operations. The spec should define **quarantine blast radius** — what other modules are affected when a specific module is quarantined, and what degraded-mode behavior activates for each quarantine scenario. Without this, a single quarantine event can produce an undefined system state.

6. Commentary on Failure Taxonomy

6.1 Seam Detection Methodology

COPILOT

The "feel → test → prove → fix" loop is unusual in formal engineering but aligns well with exploratory testing methodologies. For production systems, this should be supplemented with **automated seam detection** — scheduled probing tests that hit known seam points. Each known failure class should have a corresponding probe that runs on a cadence and reports results to the monitoring infrastructure.

GROK

The failure taxonomy is one of the strongest sections of the spec because it classifies failures by **structural cause**, not by symptom. This is the correct engineering approach. The six failure classes (Reinjection, Near-Duplicate Bypass, Fragment Survival, Context Echo Contamination, Budget Starvation, Cross-Layer Contamination) form a reasonably complete basis for boundary failures in multi-agent systems. The classification is well-partitioned — the classes are mutually exclusive and collectively exhaustive for the failure space they address.

CLAUDE

The multi-angle probing methodology — "hit the same seam from a different angle to confirm it's structural" — is a **genuine contribution to AI systems engineering**. Most failure detection in AI systems is symptom-based. This approach is cause-based and structural. The key insight: *"A seam is more believable when it survives a change in angle."* This should be elevated from a seam report to a **formal testing doctrine** — a prescribed methodology for validating any suspected structural failure in the system.

✓ Convergence Note

All three models recognize the failure taxonomy as **unusually strong and practically useful**. The cause-based classification approach and multi-angle probing methodology receive unanimous endorsement as best-in-class for multi-agent system failure analysis.

6.2 Engineering Laws

COPILOT

"One identity, one canonical instance" is the most enforceable of the five laws. The others require contextual judgment that may be difficult to automate. Enforcement strategy: implement the most enforceable laws first as hard constraints, then layer the judgment-dependent laws as soft constraints with operator-facing alerts.

GROK

"Incomplete input is invalid input" needs qualification. In probabilistic systems, **all input is incomplete** in some sense. The law should specify what constitutes "complete" input for each module — essentially, a required-fields contract. Without a module-specific completeness definition, the law is either vacuously true (all input is invalid) or unenforceable (no clear test for completeness).

CLAUDE

"Output is not input unless explicitly allowed" is the law with the **highest protective value**. It prevents the most insidious class of self-referential failures. But it's also the hardest to enforce in practice because LLM-based systems inherently treat prior output as input context. The law essentially requires the system to draw a hard line between "output that becomes context" (allowed, governed) and "output that becomes input" (prohibited by default). The spec should define precisely where that line falls in each module.

7. Commentary on Security Architecture

7.1 Policy Engine Design

COPILLOT

The single-function policy check (`check_action(caller, resource, action) → Decision`) is elegant and extensible. Rust as the enforcement language is the right choice for performance-critical policy evaluation. The Python orchestration layer calling into Rust policy is a clean separation of concerns — Python handles complexity, Rust handles speed and memory safety.

GROK

The 1–12 sensitivity scale needs **formal calibration**. What distinguishes sensitivity 7 from sensitivity 8? Without a calibration guide, different operators will assign sensitivity levels inconsistently. The spec should include a sensitivity rubric with examples for each level, analogous to how data classification schemes define each tier with concrete criteria and representative data types.

CLAUDE

The Resource Type enum is well-designed but missing a critical type: **"Identity."** Identity itself should be a protected resource. Reading a subsystem's identity manifest, modifying identity parameters, or impersonating another subsystem's identity should all be policy-gated actions. Currently, identity governance is handled by the Cognitive Bridge Module, but it should *also* be visible to the security layer. Identity is both a governance concept and a security surface —

treating it as only the former leaves a gap.

⚠ **Tension Note**

Claude identifies a **missing security surface** (Identity as a resource type) that neither Copilot nor Grok flagged. This is a significant finding — identity operations are currently governed but not security-gated, creating a potential bypass path.

8. Commentary on Core Invariants

8.1 Identity Stability

COPILLOT

The four identity stability invariants are clear and enforceable. The key implementation challenge is detecting "adopted operator sovereignty" in Nova — this requires **semantic analysis** of Nova's output, not just structural checks. Pattern matching on authority-claiming language is a start but insufficient for subtle cases where Nova implicitly assumes governance authority through framing rather than explicit claims.

GROK

These invariants should be expressed as **formal logical predicates** that can be checked automatically. For example: $\forall \text{turn } t: \text{nova_output}(t) \not\in \text{operator_governance_vocabulary}$. This makes the invariant machine-checkable rather than human-interpretable. Each invariant should have a corresponding

predicate, a test harness, and a violation-reporting mechanism.

CLAUDE

The identity stability invariants are necessary but not sufficient. They define what each subsystem **must NOT do** but don't fully define what each subsystem **MUST do**. Negative constraints prevent drift but don't ensure convergence to the correct behavior. The spec should add **positive identity invariants** — "Jarvis MUST respond with operator-grade authority when governance is required" alongside "Nova MUST NOT assume operator sovereignty." Without positive invariants, a subsystem could satisfy all negative constraints by doing nothing.

8.2 Truth Preservation

COPILLOT

"Runtime truth precedes presentation" is the right ordering. Implementation note: this means the system must always **compute the true answer before applying any mythic framing** or narrative presentation. The presentation layer is strictly post-processing. The implementation should enforce this through pipeline ordering — truth computation completes and commits before the presentation stage begins. No parallel execution of truth and presentation.

GROK

Truth preservation invariants assume a **single canonical truth**. In a multi-agent system with probabilistic inference, "truth" may be a distribution, not a point. The invariants should specify what counts as "verified state" — is it the LLM's highest-probability output, the God Brain's routed selection, or Jarvis's

approved meaning? Without a truth-selection function, the invariant is ambiguous across the system's probabilistic components.

CLAUDE

The deepest risk to truth preservation is not mythic override (which the spec guards against well) but **truth erosion** — cases where the system technically preserves truth but presents it in a way that systematically biases operator interpretation. The mythic translation layer is particularly vulnerable to this: a mythic mapping that is technically accurate but emotionally misleading could pass all invariant checks while still corrupting operator understanding. Truth preservation must extend beyond factual accuracy to **interpretive fidelity**.

9. Open Questions & Unresolved Design Tensions

The following questions emerged from tri-model review and remain unresolved as of this codex version. Each represents a design decision that the spec must eventually address.

#	Question	Description	Raised By
9.1	Meta-Governance Recursion	Who governs the governance layer? The Cognitive Bridge Module governs subsystem behavior, but what governs the Cognitive Bridge Module? The spec should define the meta-governance hierarchy	Claude (primary), Grok (supporting)

#	Question	Description	Raised By
		explicitly, including termination conditions that prevent infinite recursion.	
9.2	Temporal Scope of Governance	Most governance in the spec is turn-scoped. But identity drift, doctrine erosion, and behavioral pattern changes are trajectory-scoped. The spec needs a temporal governance layer that operates across sessions, with accumulation windows and decay functions.	Claude (primary), Copilot (supporting)
9.3	Graceful Degradation Under Partial Failure	If the Cognitive Bridge Module itself fails, what happens? The spec defines failure modes for other modules but doesn't define the degradation path when the translation layer is unavailable. Without the bridge, can the system operate in a reduced-capability mode?	All three models
9.4	Operator Error Handling	The spec assumes a competent operator. What happens when the operator issues contradictory commands, misuses mythic terminology, or	Claude (primary), Copilot (supporting)

#	Question	Description	Raised By
		triggers unintended doctrine bindings? The system needs an operator-error detection and correction path that preserves sovereignty while preventing damage.	
9.5	Scalability of Mythic Translation Registry	As the system grows, the registry will grow. At what point does registry lookup become a performance bottleneck? Should the registry support hierarchical or categorical lookups? What is the governance overhead for adding new entries at scale?	Copilot (primary), Claude (supporting)
9.6	Cross-Model Formalization Gap	Grok's formalization approach (geometric, mathematical) and Claude's logic-pressure approach (temporal, philosophical) address different aspects of the same problems. The spec should integrate both — formal correctness and philosophical coherence are both necessary for a system of this ambition.	All three models (meta-observation)

10. Recommendations by Model

10.1 Copilot Recommendations

Engineering Pragmatics & Implementation Readiness

1. **Packet Schema Versioning** — Add schema version fields and version-negotiation logic to the Direct Internal Pipeline from day one. Schema evolution without versioning will produce silent failures.
2. **Observability Infrastructure** — Build monitoring and observability alongside governance infrastructure. Every governance decision should emit a metric. Dashboards should visualize traffic class distribution, drift scores, and immune protocol response rates in real time.
3. **Automated Seam Probing** — Implement scheduled probing tests for each of the six known failure classes. Run them on a cadence. Track regression over time.
4. **Sensitivity Calibration Rubric** — Add a calibration guide for the 1-12 sensitivity scale with concrete examples at each level to prevent inconsistent operator classification.
5. **Escalation Thresholds** — Define explicit, parameterized escalation logic for Immune Protocol responses, including time-based and severity-based triggers for moving between ALLOW, CLAMP, REROUTE, REJECT, and QUARANTINE.

10.2 Grok Recommendations

Formalization & Mathematical Rigor

6. **Delivery Semantics** — Formalize delivery semantics for the internal pipeline: ordering guarantees (FIFO or otherwise), delivery mode (at-most-once, at-least-once, exactly-once), retry logic, and acknowledgment protocol.
7. **Machine-Checkable Invariants** — Express all core invariants as formal logical predicates with corresponding test harnesses. Every invariant should have a predicate, a checker, and a violation reporter.

8. **Geometric Drift Functions** — Define drift as a distance function over behavior manifolds, making drift detection a geometric problem with measurable convergence and threshold properties.
9. **Immune Protocol State Machine** — Formally specify the state machine for Immune Protocol classifications, including all valid transitions, transition triggers, and forbidden state changes.
10. **Traffic Class Type System** — Define traffic classes as a compile-time type system where possible, preventing class misassignment at the structural level rather than catching it at runtime.

10.3 Claude Recommendations

Logic-Pressure Testing & Governance Completeness

11. **Meta-Governance Recursion** — Address the meta-governance problem explicitly. Define who governs the governance layer and provide a formal termination condition for the recursion.
12. **Temporal Governance** — Add session-spanning and trajectory-spanning drift detection. Per-turn monitoring alone cannot catch slow drift. Implement accumulation windows with configurable decay.
13. **Quarantine Blast Radius** — For each quarantinable module, define the downstream impact, the degraded-mode behavior, and the recovery path. No quarantine should produce an undefined system state.
14. **Positive Identity Invariants** — Add invariants that define what each subsystem **MUST** do, not just what it must **NOT** do. Negative constraints alone allow vacuous compliance.
15. **Identity as ResourceType** — Add "Identity" as a protected resource in the security layer's ResourceType enum. Identity operations should be policy-gated, not just governance-monitored.

16. **Formal Testing Doctrine** — Elevate multi-angle seam probing from a reporting methodology to a formal testing doctrine with prescribed steps, required evidence thresholds, and standardized reporting formats.

Appendix A: Model Attribution Key

Code	Model	Analytical Role
[CPT]	Copilot (Microsoft)	Engineering pragmatics, integration design, implementation feasibility, observability architecture, and production-readiness assessment
[GRK]	Grok (xAI)	Formalization, mathematical rigor, type-theoretic analysis, edge-case pressure-testing, and formal verification methodology
[CLD]	Claude (Anthropic)	Logic-pressure testing, philosophical coherence, failure-mode completeness, governance recursion analysis, and temporal governance design

Appendix B: Commentary Cross-Reference to AAIS_MASTER_SPEC v1.2

Codex Section	Codex Topic	MASTER_SPEC Section	MASTER_SPEC Topic
§2.1	Governed Cognition Principle	§1 — Philosophy	System Philosophy & Core Principles
§2.2	Operator Sovereignty	§1 — Philosophy	Operator Authority Model
§3.1	Direct Internal Pipeline	§2 — Architecture	Internal Pipeline & Packet Model
§3.2	Lane Separation & Traffic Classes	§2 — Architecture	Traffic Classification & Lane Routing
§4.1	Intent Resolution	§3 — Cognitive Bridge	Intent Resolver & GIP Schema
§4.2	Mythic Translation Registry	§3 — Cognitive Bridge	Mythic Translation Layer
§4.3	Drift Detection	§3 — Cognitive Bridge	Identity Drift Monitor
§5.1-5.2	Immune Protocol	§4 — Immune System	Immune Protocol & Response Types
§6.1-6.2	Failure Taxonomy	§5 — Failure Analysis	Failure Classes & Engineering Laws
§7.1	Policy Engine Design	§6 — Security	Security Architecture & Policy Engine
§8.1-8.2	Core Invariants	§7 — Invariants	Identity Stability & Truth Preservation

Appendix C: Convergence & Tension Summary

C.1 Convergence Points

ID	Domain	Convergence Description	Status
C-01	System Philosophy	"Governed cognition over autonomous improvisation" is the correct foundational principle for AAIS	Confirmed
C-02	Architecture	Fast-lane/slow-lane separation and the packet model are the correct abstractions for internal communication	Confirmed
C-03	Cognitive Bridge	Intent Resolution is the highest-risk, highest-impact subsystem in the Cognitive Bridge Module	Confirmed
C-04	Cognitive Bridge	Mythic Translation Registry is architecturally necessary and well-conceived as a controlled-vocabulary pattern	Confirmed
C-05	Failure Taxonomy	The failure taxonomy is unusually strong, with cause-based classification and multi-angle probing both recognized as best-in-class	Confirmed

C.2 Tension Points

ID	Domain	Tension Description	Models	Status
T-01	Governance-Cognition Boundary	Can the boundary be made formally clean (Grok) or does it inherently require operator judgment (Claude)?	GRK vs. CLD	Unresolved
T-02	Pipeline Delivery Semantics	Missing delivery semantics identified as critical gap; disagreement on whether to prioritize versioning, delivery guarantees, or prose-boundary definition first	All three	Unresolved
T-03	Mythic Registry Completeness	Can operational concepts exist without mythic mappings? Inverse completeness question raised by Claude, unaddressed by others	CLD (solo)	Unresolved
T-04	Drift Detection Abstraction	Three-way divergence: geometric distance (Grok) vs. temporal accumulation (Claude) vs. quantitative scoring	All three	Unresolved — potentially complementary

ID	Domain	Tension Description	Models	Status
		(Copilot)		
T-05	Identity as Security Surface	Identity not included in ResourceType enum; identified by Claude as a missing security surface that creates a potential bypass path	CLD (solo)	Unresolved
T-06	Positive vs. Negative Invariants	Current invariants are exclusively negative (must NOT). Claude argues positive invariants (MUST) are needed to prevent vacuous compliance	CLD (primary)	Unresolved

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